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①

① A rigid spacecraft has principal moments of inertia $(1000, 1400, 1700) \text{ kg-m}^2$. This spacecraft is placed into circular orbit at altitude of 500 km above Earth. $R = 6400 \text{ km}$.

② How should three principal axes of the spacecraft be aligned with respect to LVLH so that the spacecraft is stable in pitch, roll, yaw according to linearized equations for rotational motion?

③ Corresponding to the conditions you obtained in part ②, simulate nonlinear on orbit spacecraft rotational eq to determine the Euler angle time response, assuming that each initial Euler angle is 0.1 rad and Euler rate ratios compare these time response with your results in part ②.

④ Corresponding to the conditions you obtained in part ②, what are the eigenvalue for pitch, roll, yaw dynamics?

①-a $I_y > I_x > I_z$ 이면 stable 하므로, $1700 > 1400 > 1000$ $I_y = 1700, I_x = 1400, I_z = 1000$ 으로 잡자.

①-b 선형화 된 중. 피. 회 식별 노트 30 쪽에서 space state form으로 가져오자. $\omega = \frac{v}{R} = \frac{4000}{6400} = 0.001 \text{ rad/s}$

①-1
$$\begin{bmatrix} \Delta \dot{\theta} \\ \Delta \dot{w}_y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{3\omega^2(I_z - I_x)}{I_y} & 0 \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta w_y \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{I_y} \end{bmatrix} M_y$$
 , eigenvalue $= \det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -8.541176 \times 10^{-9} & -\lambda \end{vmatrix} = \lambda^2 - 8.541176 \times 10^{-9} = 0$
 $\lambda = \pm 0.93j$

①-2
$$\begin{bmatrix} \Delta \dot{\phi} \\ \Delta \dot{w}_x \\ \Delta \dot{\psi} \\ \Delta \dot{w}_z \end{bmatrix} = \begin{bmatrix} 0 & 1 & -\omega & 0 \\ \frac{3\omega^2(I_z - I_y)}{I_x} & 0 & 0 & \frac{\omega(I_y - I_z)}{I_x} \\ \omega & 0 & 0 & 1 \\ 0 & \frac{\omega(I_x - I_y)}{I_z} & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta w_x \\ \Delta \psi \\ \Delta w_z \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{I_x} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I_z} \end{bmatrix} \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$
 , eigenvalue $= \det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & -\omega & 0 \\ \frac{3\omega^2(I_z - I_y)}{I_x} & -\lambda & 0 & \frac{\omega(I_y - I_z)}{I_x} \\ \omega & 0 & -\lambda & 1 \\ 0 & \frac{\omega(I_x - I_y)}{I_z} & 0 & -\lambda \end{vmatrix} = 0$
 $\lambda = \pm 0.0017j, \pm 0.00055j$ 안정.

①-c
$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \phi_0 \\ \theta_0 \\ \psi_0 \end{pmatrix} = \begin{pmatrix} 0 & \sin \theta_0 \cos \phi_0 & \cos \theta_0 \cos \phi_0 \\ 0 & \cos \theta_0 & -\sin \theta_0 \cos \phi_0 \\ 1 & \sin \theta_0 \sin \phi_0 & \cos \theta_0 \sin \phi_0 \end{pmatrix} \begin{pmatrix} w_{x0} \\ w_{y0} \\ w_{z0} \end{pmatrix}$$

$$\begin{pmatrix} w_{x0} \\ w_{y0} \\ w_{z0} \end{pmatrix} = \begin{pmatrix} w_{x0} \\ w_{y0} \\ w_{z0} \end{pmatrix} + \begin{pmatrix} w_{x0} \cos \theta_0 \sin \phi_0 \\ w_{x0} \cos \theta_0 \cos \phi_0 + w_{y0} \sin \theta_0 \sin \phi_0 \\ w_{x0} \sin \theta_0 \cos \phi_0 - w_{y0} \sin \theta_0 \sin \phi_0 \end{pmatrix} = \begin{pmatrix} 0.001 \times 0.0015 \times 0.0008 \\ 0.001 \times 0.0015 \times 0.0015 + 0.001 \times 0.0008 \times 0.0008 \times 0.0008 \\ 0.001 \times 0.0015 \times 0.0008 - 0.001 \times 0.0008 \times 0.0008 \times 0.0008 \end{pmatrix} = \begin{pmatrix} 0.0001 \\ 0.0001 \\ 0.0001 \end{pmatrix}$$

결과론 초기 요잉각과 초기 각속도가 평형점과 거의 동일하게 나온다.

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} w_x \\ w_y \\ w_z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

② A rigid spacecraft has principal moments of inertia $I_x=1200$, $I_y=1000$, $I_z=800$ (kg-m^2) defined with R.R.T. principal axes. This spacecraft is placed into a geosynchronous circular orbit around Earth so that $x\hat{x}$ is aligned with local horizontal direction in orbital plane, $y\hat{y}$ is aligned normal to orbital plane, $z\hat{z}$ is aligned along radial direction.

- What are linearized pitch eq. and their eigen value?
- What are linearized roll/yaw eq and their eigen value?
- Simulate nonlinear on-orbit spacecraft rotational eqs to determine Euler angle time responses, assuming that initial Euler angles are $\phi=0.1$, $\theta=0.1$, $\psi=0$ (rad), and the Euler angle rates are all initially zero. Compare these time responses with your results in part (b).

2-a $I_y > I_z > I_x \rightarrow$ 안정인데, 위상량론 $I_x > I_y > I_z$ 라서 불안정이다.

2-b 선형화된 롤 피치 식을 노트 30 쪽에서 state form으로 가져오자. $\omega = \sqrt{\frac{M}{R^3}} = \sqrt{\frac{400,000}{4200^2}} = 7.3 \times 10^{-5}$ (rad/s)

III $\begin{pmatrix} \Delta \dot{\theta} \\ \Delta \dot{\psi}_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{3\omega^2(I_z - I_x)}{I_y} & 0 \end{pmatrix} \begin{pmatrix} \Delta \theta \\ \Delta \psi_y \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{I_y} M_y \end{pmatrix}$, eigenvalue = $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -6.35 \times 10^{-9} & -\lambda \end{vmatrix} = \lambda^2 - 6.35 \times 10^{-9} = 0$, $\lambda = \pm 7.9 \times 10^{-5} j$

IV $\begin{pmatrix} \Delta \dot{\phi} \\ \Delta \dot{\psi}_x \\ \Delta \dot{\psi}_z \end{pmatrix} = \begin{pmatrix} 0 & 1 & -\omega & 0 \\ \frac{3\omega^2(I_z - I_y)}{I_x} & 0 & 0 & \frac{\omega(I_y - I_z)}{I_x} \\ \omega & 0 & 0 & 1 \\ 0 & \frac{\omega(I_x - I_z)}{I_z} & 0 & 0 \end{pmatrix} \begin{pmatrix} \Delta \phi \\ \Delta \psi_x \\ \Delta \psi_z \\ \Delta \psi_y \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \frac{1}{I_x} & 0 \\ 0 & 0 \\ 0 & \frac{1}{I_z} \end{pmatrix} \begin{pmatrix} M_x \\ M_z \end{pmatrix}$, eigenvalue = $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 & -7.3 \times 10^{-5} & 0 \\ -2.6 \times 10^{-6} & -\lambda & 0 & 1.2 \times 10^{-5} \\ 7.3 \times 10^{-5} & 0 & -\lambda & 1 \\ 0 & 1.5 \times 10^{-5} & 0 & -\lambda \end{vmatrix} = 0$

$\lambda = \pm 9.1 \times 10^{-5} j, \pm 2.3 \times 10^{-5}$

불안정.

2-c 노트 29 쪽

$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \dot{\phi}_0 \\ \dot{\theta}_0 \\ \dot{\psi}_0 \end{pmatrix} = \begin{pmatrix} 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \\ 0 & \cos \phi & -\sin \phi \\ 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \end{pmatrix} \begin{pmatrix} \omega_{x0} r \\ \omega_{y0} r \\ \omega_{z0} r \end{pmatrix} = \begin{pmatrix} 0 & 0.0048 \times 1.005 & 0.9945 \times 1.005 \\ 0 & 0.9945 & -0.0048 \\ 1 & 0.0048 \times 0.1003 & 0.9945 \times 0.1003 \end{pmatrix} \begin{pmatrix} \omega_{x0} r \\ \omega_{y0} r \\ \omega_{z0} r \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} \omega_{x0} r \\ \omega_{y0} r \\ \omega_{z0} r \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$\phi_0 = 0$
 $\theta_0 = 0.1$
 $\psi_0 = 0.1$

$\begin{pmatrix} \omega_{x0} \\ \omega_{y0} \\ \omega_{z0} \end{pmatrix} = \begin{pmatrix} \omega_{x0} r \\ \omega_{y0} r \\ \omega_{z0} r \end{pmatrix} + \begin{pmatrix} \omega_x \cos \phi \sin \theta \\ \omega_x \cos \phi \cos \theta + \omega_y \sin \phi \sin \theta \\ \omega_y \cos \phi \sin \theta - \omega_x \sin \phi \cos \theta \end{pmatrix} = \begin{pmatrix} 0 \\ 7.3 \times 10^{-5} \times 0.9945 \times 0.9945 \\ -7.3 \times 10^{-5} \times 0.0048 \times 0.9945 \end{pmatrix} = \begin{pmatrix} 0 \\ 7.2 \times 10^{-5} \\ 7.2 \times 10^{-6} \end{pmatrix}$

☆ 결과로는 초기 오일러각과 초기 각속도가 평형점과 다르게 나온다.

1-C 와 2-C 그래프이다.

(3)

1-C 는 위성자세가 안정이다. 비선형으로 검사해보니 위성자세가 안정하고 주기값의 경향이 비슷하다.
2-C 는 위성자세가 불안정이다. 비선형으로 검사해보니 위성자세가 안정하지만 주기값 경향이 안-보인다.

